

2026-06-07-helix-ota-design

Helix OTA — Master Design Specification

Field	Value
Revision	1
Created	2026-06-07
Last modified	2026-06-07
Status	active (design — pending operator review gate)
Status summary	Canonical master design for the Helix OTA universal over-the-air update system. Anchors the foundation corpus: vision, locked decisions, architecture, mandated stack, 1.0.0-MVP scope, security/trust model, data model, rollout engine, telemetry, submodule reuse + new-repo map, phased directory layout, documentation/export pipeline, four-layer testing strategy, and the multi-agent execution model. Synthesizes the two operator drafts via <u>../research/additions_synthesis.md</u> .
Issues	Wrapped-engine, trust-framework, and topology choices are deliberately open pending evidence-based ADRs (ADR-0001..0005).
Issues summary	No architectural commitment is made without research evidence (§11.4.6/§11.4.8).
Fixed	initial master design
Fixed summary	Foundation-corpus-first design produced from operator brief + HelixConstitution + submodules catalogue + additions drafts.
Continuation	On approval: write per-component 1.0.0-MVP specs, run the research

Field	Value
	ADRs, create the new submodule repos, then dispatch authoring + code-review workflows; render multi-format exports on stabilized content.

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§1. Vision, scope, and non-goals

Vision. A universal, generic, deeply decoupled OTA system — server control plane + per-OS client SDKs/agents + dashboard — embeddable into any operating system. First target: **Android 15 (all flavors/variants) on Orange Pi 5 Max**, where the build pipeline emits flashing images + OTA update .zip + mandatory hash files. Later phases extend to Linux (all flavors), Windows, and other upstream OSes via pluggable OS adapters.

Hard guarantees (operator-stated). - Zero system corruption: an update must never brick or break a working device. - Safe upload: every OTA artifact passes mandatory validation before it can be deployed. - Granular rollout: deploy to all at once, or in steps (5%, 10%, 30%, ... 100%). - Observability: tracking/measurement/critical-data capture to detect and report problems. - Scalability: single board → millions of devices.

Non-goals for 1.0.0-MVP. End-user/multi-version rollback (deferred; automatic A/B boot-failure rollback IS included); full TUF/Uptane (deferred to 1.0.1+); non-Android OS clients; delta updates; microservice decomposition.

§2. Locked decisions

#	Decision	Source
D1	Foundation corpus first, then go wide (research → MVP spec → review gate → phase depth).	Operator
D2	Device-side native Android A/B (update_engine + AVB/dm-verity + auto-rollback) + custom decoupled Go control plane; wrap an OSS engine only where it adds value.	Operator
D3	The wrapped-engine choice (hawkBit / Mender / AOSP-native-only) is decided by the evidence-based research ADR, not pre-committed.	Operator + §11.4.8
D4	New reusable submodules get PUBLIC repos auto-created on GitHub + GitLab (listed before bulk creation).	Operator
D5	additions/ files are authoritative input — deeply analyzed and folded in.	Operator
D6	Mandated stack: Go + Gin + Brotli + HTTP/3(QUIC)→HTTP/2 fallback, REST primary + all compatibility interfaces.	Operator
D7	MVP defaults: 15 min + jitter poll (configurable); React dashboard; PostgreSQL + MinIO; OpenTelemetry.	Default (overridable)

§3. Mandated technology stack

- **Language/runtime:** Go (control plane, rollout engine, validators), Kotlin/KMP (Android agent).
- **HTTP framework:** Gin (gin-gonic).

- **Transport: HTTP/3 (QUIC)** primary via the vasic-digital/http3 submodule (drop-in net/http.Handler), automatic **fallback to HTTP/2 + standard compression; Brotli** content compression with negotiated fallback (gzip) for older clients.
- **API surface: REST** is the mandated primary surface (/api/v1), plus all mandatory compatibility interfaces; **gRPC optional/internal only**.
- **Persistence:** PostgreSQL (relational), MinIO/S3 (artifact blobs), cache brick (optional Redis only if needed).
- **Observability:** OpenTelemetry via the observability brick; Prometheus/Grafana surface.
- **Containerization:** the vasic-digital/containers submodule is the canonical substrate (§11.4.76).

§4. System architecture

Three planes plus infra, with two deliberately extractable seams (the **OS-adapter** seam for universality and the **rollout-engine** seam for OS-agnostic campaigns).

```
graph TB
    subgraph Dashboard ["Dashboard (React)"]
        UI["Helix OTA Dashboard – secure login, upload, rollout, fleet health"]
    end
    subgraph Control ["Control Plane (Go + Gin, HTTP/3→HTTP/2, Brotli)"]
        API["REST API /api/v1 (+ optional internal gRPC)"]
        ART["Artifact intake + validation + signing-verify"]
        ROLL["Rollout engine (staged %, halt-on-failure)"]
        DEV["Device registry + inventory"]
        TEL["Telemetry ingest + health"]
        AUTH["Auth (OAuth2/JWT, RBAC)"]
    end
    subgraph Data ["Data plane"]
        PG[(PostgreSQL)]
        OBJ[(MinIO / S3)]
        OTEL[(OpenTelemetry / Prometheus)]
    end
    subgraph Device ["Android 15 device (Orange Pi 5 Max)"]
        AGENT["OTA agent (KMP) – poll, download, verify"]
        UE["update_engine (Virtual A/B + compression)"]
        AVB["AVB / dm-verity + boot_control"]
    end
    UI --> API
    API --> ART --> OBJ
    API --> ROLL --> PG
    API --> DEV --> PG
    API --> TEL --> OTEL
    API --> AUTH
    AGENT -->|HTTPS poll/report| API
    AGENT --> UE --> AVB
```

Decoupling principle (§11.4.28). Each unit has one purpose, a well-defined interface, and is independently testable: artifact-validator, rollout-engine, ota-protocol types, the update_engine bridge, and the OS-adapter are separate modules so future OSes and future projects can reuse them.

§5. 1.0.0-MVP definition

End-to-end happy path: **admin logs in → uploads signed OTA .zip + hash → server validates (structure, hash, signature, version monotonicity, target compatibility) → publishes release → deploys (all-at-once for MVP; staged engine lands 1.0.1) → device polls (15 min + jitter), downloads, re-verifies, applies via update_engine to the inactive slot, reboots → update_verifier confirms → telemetry success/failure with automatic A/B rollback on boot failure.**

MVP components: Auth, Artifact intake+validation, Device registry, Release/Deploy (all-targets), Telemetry ingest, Dashboard (login/upload/deploy/health), Android agent, containerized dev stack.

§6. Security & trust model

- **Artifact signing:** build-pipeline private key signs; public key in the device trust store; server verifies on upload, device re-verifies before apply.
- **Integrity:** SHA-256 (and SHA-512 where available) over the artifact + the mandatory hash file.
- **Transport:** TLS 1.3; HTTP/3(QUIC) with HTTP/2 fallback.
- **Device identity:** token bound to hardware id (Android KeyStore); mutual-TLS evaluated.
- **Anti-corruption / anti-downgrade:** AVB + dm-verity + A/B boot_control; bootloader version checks.
- **Audit:** every admin action logged.
- **Forward path:** TUF/Uptane is the 1.0.1+ hardening (ADR-0002); signing interfaces are designed so Uptane drops in without rework.

§7. Data model overview

Base schema synthesized from both drafts (02 as the richer base), normalized: users, api_keys, devices, device_groups(+members), artifacts(+versions), releases, deployments, deployment_phases, device_deployments, rollouts, telemetry_events/update_metrics, audit_logs, rollback_history (1.0.1+). Full DDL + up/down migrations land in 1.0.0-mvp/database/. Indexing per draft 02 §4 plus query-driven additions.

§8. Staged rollout engine

OS-agnostic, extractable submodule. Config = ordered phases {percentage, success_threshold, error_threshold, duration, auto_progress}; engine starts a phase, targets the phase cohort, monitors telemetry, **halts/pauses on error-**

threshold breach, advances on success-threshold within duration. Seeded by draft 02 §6; the Go loop there is a design reference, hardened with deterministic cohort selection and idempotent state transitions. Lands in 1.0.1-staged-rollout/.

§9. Telemetry & observability

Device event stream (download_started/installing/installed/verifying/success/failure + error codes + system health) → ingest → OpenTelemetry/Prometheus → dashboard health + alerting via Herald. Metrics drive the rollout halt logic (§8) and problem detection/reporting (operator requirement).

§10. Submodule reuse + new repositories

Reuse/extend (verified catalogue, §11.4.74): auth, security, database, Storage, observability, eventbus, ratelimiter, middleware, http3, mdns, recovery, Herald, config, discovery, cache, docs_chain/Document/Formatters (export), containers (infra); Android: Auth-KMP, Security-KMP, Storage-KMP, Config-KMP; dashboard: UI-Components-React, Dashboard-Analytics-React, Auth-Context-React.

New submodules (PUBLIC repos to be created on GitHub + GitLab per D4):

Repo	Purpose	Decoupled boundary
ota-protocol	Shared wire types, manifest schema, status/event enums (Go + KMP).	No business logic; pure contracts.
ota-artifact-validator	Structure/hash/signature/metadata validation pipeline.	OS-aware via plugins; no transport.
ota-rollout-engine	Staged-rollout + halt/advance logic, OS-agnostic.	No HTTP; pure engine + storage port.
ota-update-engine-bridge	Wrapper over AOSP update_engine / boot_control.	Android-only; thin, testable.
ota-android-agent	KMP device agent (poll/download/verify/apply/report).	Consumes protocol + bridge.
ota-telemetry-schema	Telemetry event/metric schema + codecs.	Shared by server + agents.

(Final list confirmed in the MVP spec immediately before creation.)

§11. Phasing & directory layout

```
docs/research/main_specs/  
├─ 00-master/      this design, ADRs, glossary, threat model,  
submodule map  
├─ research/       additions_synthesis.md,  
ota_landscape_report.md, ADR-000x  
├─ additions/      operator-supplied inputs (authoritative;  
analyzed in research/)  
├─ 1.0.0-mvp/      architecture, api, database, security, server,  
client_android, tests, deployment, diagrams  
├─ 1.0.1-staged-rollout/ rollout engine, monitoring, end-user  
rollback + TUF/Uptane (1.0.1+)  
├─ 1.X-linux/ 1.X-windows/ 1.X-other-os/ future-OS research +  
standards survey  
└─ _exports/       generated PDF/HTML/DOCX + diagram renders  
(built from stable source)
```

§12. Documentation & multi-format export

Canonical source = Markdown + Mermaid. A containerized export pipeline (pandoc + mermaid-cli + drawio CLI) renders **PDF/HTML/DOCX** and **mermaid**→**svg/png/draw.io/uml** into `_exports/`, run on reviewed/stable content (reproducible at any tag). Every doc carries the Constitution metadata table + ToC (§11.4.61). Prefer reusing `docs_chain/Document/Formatters` for the pipeline.

§13. Testing strategy (four-layer)

Per Constitution §1, every change ships four layers: **source-presence gate** → **artifact gate (bytes shipped)** → **runtime/integration** → **mutation meta-test (PASS→FAIL on negation)**, with no-bluff positive evidence (§7.1). Safety-critical paths (signing-verify, apply, rollout-gate) target ≥90% coverage. Device path: emulated A/B apply + a real Orange Pi 5 Max validation plan (download→verify→apply→reboot→verify; corrupt-slot→confirm A/B fallback).

§14. Execution model

Operator authorized spawning subagents → multi-agent **Workflows**: (1) parallel research agents per OTA stack → synthesis → ADRs; (2) parallel spec-authoring agents per component; (3) **mandatory code-review subagents** (§11.4.125) adversarially verifying each artifact before it's accepted; (4) multi-format export on stable content. Foundation corpus first, operator review gate, then phase-by-phase depth.

§15. Constitution compliance map

Constitution rule	How this design honors it
§1 / §1.1 four-layer + mutation	§13 testing strategy.
§7.1 no-bluff	Draft marketing stripped; evidence-only ADRs (§16).
§11.4.6 / §11.4.8	§2 D3 — engine choice deferred to research ADR.
§11.4.74 / §11.4.76	§10 catalogue-first reuse; containers substrate.
§11.4.28	§4 decoupling principle; §10 new-submodule boundaries.
§11.4.65 / §11.4.61	§12 export pipeline + metadata/ToC on every doc.
§11.4.20 / §11.4.125	§14 subagent-driven + code-review gate.
§2 / §2.1 / §3 / §4	Multi-upstream push + submodule-commit-first + tag mirroring (applied at repo creation).

§16. Architecture Decision Records (open)

ADR-0001 wrapped engine (hawkBit vs Mender vs AOSP-native-only) · ADR-0002 supply-chain trust (signing vs TUF vs Uptane + timing) · ADR-0003 server topology (modular monolith vs microservices + scale trigger) · ADR-0004 transport (HTTP/3+Brotli rollout, HTTP/2 fallback) · ADR-0005 delta updates. Each ADR is evidence-backed by the research report and reconciles the contradictions in [../research/additions_synthesis.md](#) §5.